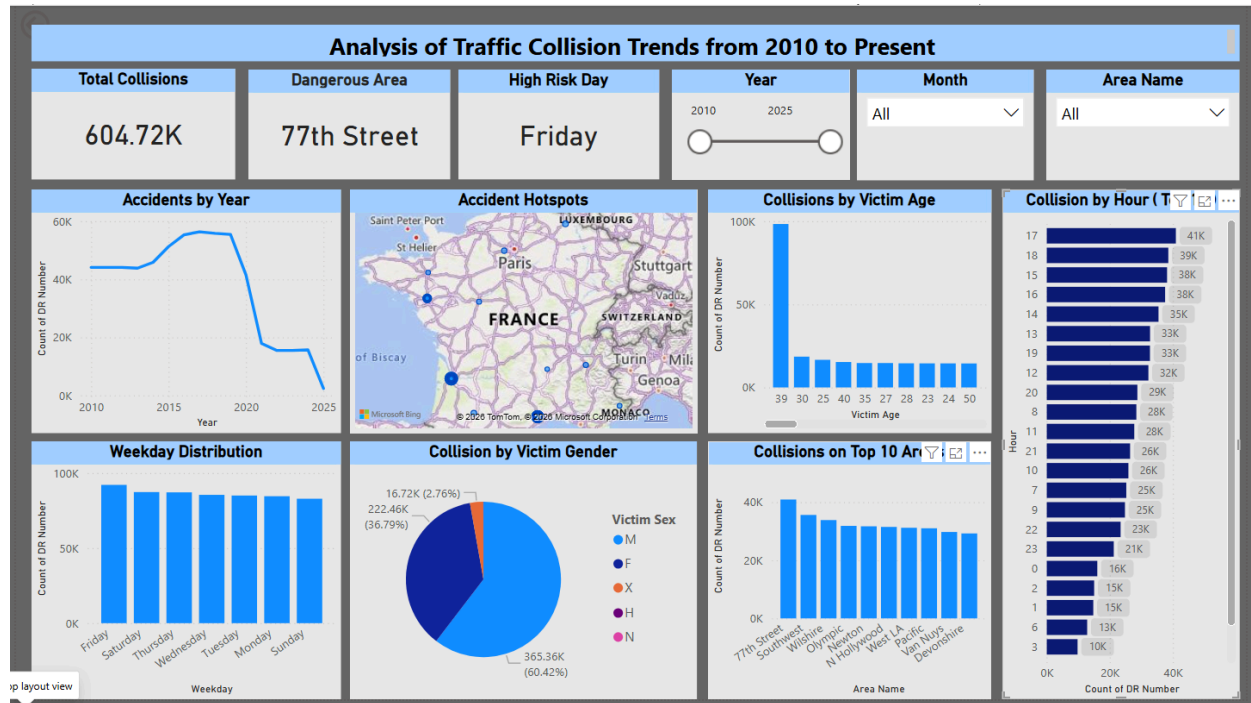


Analysis of Traffic Collision Trends from 2010 to Present



Summary

From the Stage 3 analysis, the following patterns are clearly observed:

- Traffic accidents increased steadily from 2010 to 2019, peaking around 2017–2018, and then declined sharply after 2020 due to reduced travel during COVID-19.
- Accidents are not evenly distributed throughout the year; October, January, and August show the highest accident counts, while February and November show the lowest.
- Accidents are most frequent on Fridays, followed by Saturdays and Thursdays, and least frequent on Sundays.
- A few areas, such as 77th Street, Southwest, Wilshire, Olympic, and Newton, account for a large proportion of total accidents.
- Most victims fall in the 20–50 age group, and male victims (~60%) are involved more frequently than females.

- The majority of accidents occur on streets, followed by parking lots.
- Time-of-day analysis shows a strong peak between 4 PM and 7 PM, while early morning hours (3 AM–5 AM) have the lowest accident counts.

Overall, accident risk is strongly driven by rush-hour traffic, location hotspots, and working-age commuters.

Key Insights

1. Time-Based Risk (Peak Hours)

Accidents peak between 4 PM and 7 PM, corresponding to evening rush hours.

Meaning:

Traffic congestion and commuting behavior are the main contributors to accident risk.

2. High-Risk Days

Friday has the most accidents, followed by Saturday and Thursday, while Sunday is the safest day.

Meaning:

End-of-week travel, social activities, and driver fatigue increase accident probability.

3. Location Hotspots

A small group of areas (77th Street, Southwest, Wilshire, Olympic, Newton) consistently reports the highest accident counts.

Meaning:

Accidents are geographically concentrated and not evenly spread across all regions.

4. Victim Demographics

Most victims are between 20 and 50 years old, and males account for about 60% of all cases.

Meaning:

Working-age adults with frequent commuting exposure are the most vulnerable group.

5. Seasonal Pattern

Certain months (October, January, August) show higher accident frequency than others.

Meaning:

Seasonal travel, festivals, and weather conditions influence accident rates.

Recommendations (Prescriptive Analysis)

Based on the above insights, the following actions are recommended:

1. **Peak Hour Enforcement:**

Increase traffic police deployment and monitoring between **4 PM and 7 PM**.

2. **Hotspot Area Improvement:**

Install speed cameras, improve road lighting, and redesign intersections in high-risk areas such as 77th Street and Southwest.

3. **Weekend & Friday Campaigns:**

Conduct targeted awareness and enforcement programs before weekends, especially on Fridays.

4. **Age-Focused Safety Programs:**

Promote safe-driving education for working-age adults (20–50 years), focusing on speeding and distracted driving.

5. **Seasonal Planning:**

Strengthen traffic safety measures during high-risk months like October and August.

Final Story with the Dashboard

The dashboard tells a complete story of traffic collisions by answering four key questions:

When do accidents happen?

- Mostly during **evening rush hours (4–7 PM)**
- More frequently on **Fridays and weekends**

Where do accidents happen?

- Concentrated in hotspot areas such as **77th Street and Southwest**

- Mostly on public streets and intersections

Who is most affected?

- **Working-age adults (20–50 years)**
- **Male victims** more than female victims

What does this mean?

Traffic collisions are driven by commuting behavior, congestion, and location-specific risks. The dashboard enables authorities to identify high-risk periods and locations and take preventive action.

Conclusion

This project successfully transformed raw traffic collision data into meaningful insights using data cleaning, exploratory analysis, and visualization techniques.

The dashboard highlights key trends, hotspot locations, and vulnerable population groups, providing strong support for traffic safety planning and data-driven decision-making.